

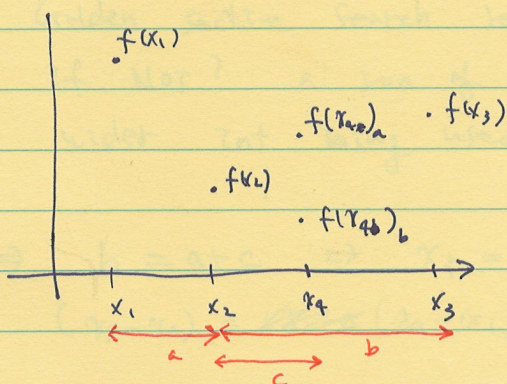
R optimize f_t^n (one-dim'l opt)

Comb. of "Golden Section Search" + "Parabolic Interpolation"
 f_t^n must be conti.

① Golden Section Search. ; Find the extremum (min or max)

of a strictly unimodal f_t^n

by succ. narrow the range
 (by Kiefer (1953))



i) Min must be in (x_1, x_3) (f is unimodal)

ii) probe the f_t^n by evaluating it at a new value of x
 , say x_4 .

iii) Choose x_4 somewhere inside the largest interval, (let $x_2 < x_3$)
 $\Rightarrow$ It is most efficient.

iv) $f(x_{4a})$ case : min $\in (x_1, x_4)$
 $\Rightarrow$ New triple pts (x_1, x_2, x_4)

$f(x_{4b})$ case : min $\in (x_2, x_3)$
 $\Rightarrow$ New triple pts (x_2, x_4, x_3)

v) Construct a new narrower search interval
 that is guaranteed to contain the f_t^n 's min.

Symbolic Math

$f = \text{expression}(x_2, 'x')$

$D(f, 'x')$ or $D(D(f, 'x'), 'x')$

deriv

⊛ probe pt selection.

New search Interval

i) $(x_1, x_4) \rightarrow \text{length} = a + c$

ii) $(x_2, x_3) \rightarrow \quad \quad \quad " = b$

Golden section Search req that these 2 int be equal.

If NOT? a run of "bad luck" could lead to the wider int being used many times $\Rightarrow$ slow conv.

$$\Rightarrow b = a + c \Rightarrow x_4 = x_1 + (x_3 - x_2)$$

$$(x_3 - x_2) = \cancel{x_2 - x_1} (x_2 - x_1)$$

What about x_2 ?

Rule: Same proportion of spacing for $(x_1, x_2, x_4) \in (x_2, x_4, x_3)$

Case 1) $f(x_{a_1})_a \rightarrow \text{subseq} = (x_1, x_2, x_4)$ orig. seq = (x_1, x_2, x_3)

$$\frac{x_2 - x_1}{a} = \frac{x_4 - x_2}{b}$$

$$a : b = c : a$$

$$\frac{a}{b} = \frac{c}{a}$$



Case 2) $f(x_{a_2})_b \rightarrow \text{subseq} = (x_2, x_4, x_3)$

$$\frac{x_4 - x_2}{b} = \frac{x_3 - x_4}{c}$$

$$c : (b - c) = a : b$$

$$\frac{c}{b - c} = \frac{a}{b}$$

$$c = \frac{a^2}{b} \text{ and } \Rightarrow \left(\frac{b}{a}\right)^2 = \frac{b}{a} + 1$$

$$\text{let } \frac{b}{a} = x, \quad x^2 - x - 1 = 0 \quad x = \frac{1 + \sqrt{5}}{2} = 1.6180$$

$$b = 1.6180 a$$

⊛ successive parabolic Interpolation.

quadratic (poly of degree two)